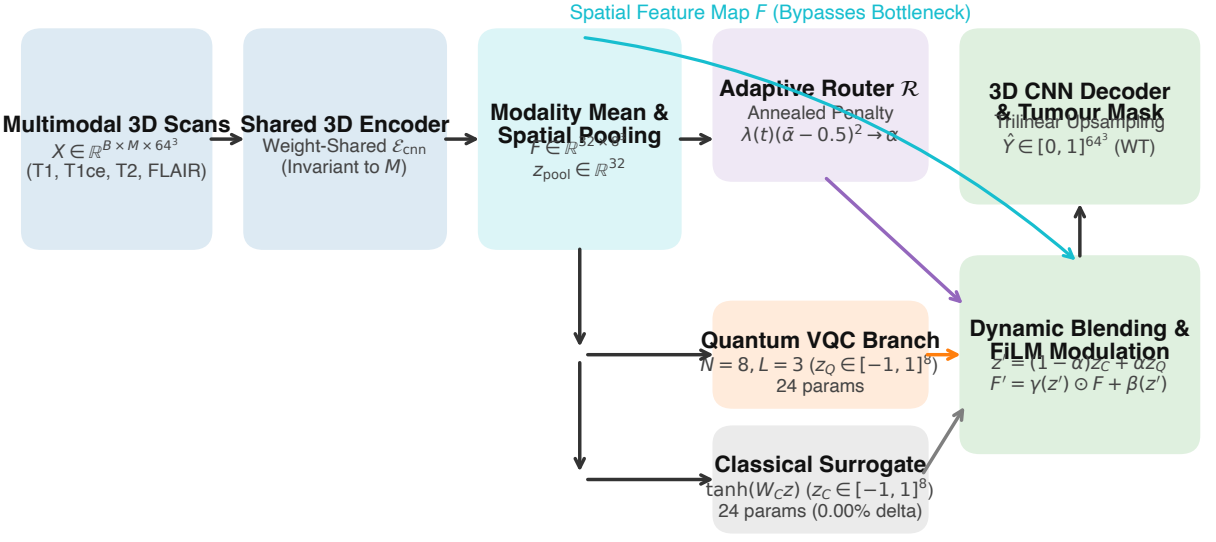
Spatial Feature Map F (Bypasses Bottleneck)
Multimodal 3D Scans
$X \in \mathbb{R}^{B \times M \times 64^3}$
(T1, T1ce, T2, FLAIR)
Shared 3D Encoder
Weight-Shared $\mathcal{E}_{\text{cnn}}$
(Invariant to M)
Modality Mean & Spatial Pooling
$F \in \mathbb{R}^{32 \times 8}$
$z_{\text{pool}} \in \mathbb{R}^{32}$
Adaptive Router $\mathcal{R}$
Annealed Penalty
$\lambda(t)(\bar{\alpha} - 0.5)^2 \to \alpha$
3D CNN Decoder & Tumour Mask
Trilinear Upsampling
$\hat{Y} \in [0, 1]^{64^3}$ (WT)
Quantum VQC Branch
$N = 8, L = 3$ ($z_Q \in [-1, 1]^8$)
24 params
Classical Surrogate
$\tanh(W_C z)$ ($z_C \in [-1, 1]^8$)
24 params (0.00% delta)
Dynamic Blending & FiLM Modulation
$z' = (1 - \alpha) z_C + \alpha z_Q$
$F' = \gamma(z') \odot F + \beta(z')$